

Lab Worksheet 5: Iteration II

CS 101 Algorithmic Problem Solving

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1. Family Legacy [1.25 Points]

In the futuristic city of Byteville, where holographic screens and AI-powered companions are the norm, a young coder named Ava stumbles upon an old data drive in her family's attic. This seemingly ancient artifact contained a cryptic Python script for creating a stunning numerical pattern. Intrigued by its simplicity and elegance, Ava set out to decipher the script and uncover the secrets behind the mesmerizing pattern.

Ava, fueled by her passion for programming, meticulously studied the script, noticing the clever use of nested loops. As she executed the code, it prompted for an integer input and Ava complied; she entered a number N and the holographic display came alive, showcasing a cascade of numbers where each row displayed a series of numbers in ascending order with each subsequent row containing less numbers. Ava showed you the pattern generated, but refused to show you its code as she believes it to be a part of her family's legacy.

You wish to recreate this program. Given an integer N , create a program that prints the pattern described above (and shown below).

Constraints

- $N \in \mathbb{Z}$
- $1 \leq N \leq 10$

Interaction

The input comprises a single line containing a single integer denoting the value of N .

The output must consist of N lines containing the pattern as described above (and shown below).

Sample

Input	Output
2	1 2 1
3	1 2 3 1 2 1

In the first case, $N = 2$. The pattern consists of 2 lines. Each line contains integers in ascending order (starting from 1) separated by a space.

In the second case, $N = 3$. The pattern consists of 3 lines. Each line contains integers in ascending order (starting from 1) separated by a space.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
4	1 2 3 4 1 2 3 1 2 1
5	1 2 3 4 5 1 2 3 4 1 2 3 1 2 1

Pseudocode

Input: N

Output:

```

N=int(input("enter number here: "))
for i in range(1,N+1):
    for j in range(1,N+1):
        print(j,end=" ")
    print()
    N=N-1

```

Dry Run

Below, dry run your pseudocode on any **one** of the inputs provided in the Exercise section.

Input: 4

<pre> N=4 for i in range(1,4+1): for j in range (1, 4+1): print(j,end="") (j= 1 2 3 4) print() (newline) n=n-1 (n=3) for j in range (1, 3+1): print(j,end="") (j= 1 2 3) print() (newline) n=n-1 (n=2) </pre>	<pre> for j in range (1, 2+1): print(j,end="") (j= 1 2) print() (newline) n=n-1 (n=1) for j in range (1, 1+1): print(j,end="") (j= 1) print() (newline) n=n-1 (n=0) </pre>	Output: <pre> 1 2 3 4 1 2 3 1 2 1 </pre>
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2. Secret Code [1.25 Points]

Ali and his town is under attack, they are waiting for help. The help arrives in form of supplies in the boxes which are labelled with a code each. Ali is skeptical of those boxes but he knows the secret code the attackers use. If the label has an odd number of vowels then it is from the attackers.

Given the label l , figure out if it is safe to open or not.

Constraints

- $1 \leq \text{len}(l) \leq 50$

Interaction

The input comprises a single line containing a string that denotes l .

The output must contain a string stating 'safe' or 'not safe' accordingly.

Sample

Input	Output
ahijklo	not safe
kljiom	safe

In the first case, the label 'ahijklo' has 3 vowels (a,i,o). Since the number of vowels is odd (3), hence it is not safe to open the box.

In the second case, the word 'kljiom' has 2 vowels. Since the number of vowels is even, the box is safe to open.

Proposed Solution

```
vowels = 0
for letter in l:
    if l in "aeiou":
        vowels+=1
if vowels % 2 != 0:
    print("safe")
else:
    print("not safe")
```

Dry Run

Using any one of the inputs provided in the Sample section above, dry run the proposed pseudocode in the space below.

```
Input: l=kljiom
Output:
vowels = 0
for letter in l:
    if l in "aeiou": (true, checks whole list at the same time)
        vowels+=1 (vowel=1)
    if vowels % 2 != 0: (true)
        print("safe")
    else:
        print("not safe")
Output: not safe (expected answer= safe)
```

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and errors in each line.

it should be if "letter in aeiou" instead of "l in aeiou". then in line 5 it should be (==) instead of (!=)

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

```
vowels = 0
l=int("enter here: ")
for i in l:
    if i in "aeiou":
        vowels+=1
    else:
        vowels=vowels
if vowels%2==0:
    print("safe")
else:
    print("not safe")
```

3. The Discovery of Dr. Khan [1.25 Points]

In the heart of Lahore, at the prestigious Lahore Institute of Science and Technology (LIST), there was a brilliant yet unconventional scientist named Dr. Ahmed Khan. Dr. Khan was renowned for his pioneering research in the field of molecular cryptography. One fateful day, while exploring the institute's archives, he stumbled upon a vial containing a three-letter string, "XYZ," which had remained a mystery for decades.

Driven by his insatiable scientific curiosity, Dr. Khan believed that deciphering this enigmatic code could unlock a groundbreaking scientific formula. He saw it as a challenge worthy of his intellect and a chance to make a significant contribution to Pakistani science.

To unravel the secrets hidden within "XYZ," Dr. Khan embarked on a daring experiment. His goal was to find all possible permutations of these three letters, hoping that one of these permutations might reveal the long-lost formula.

Suppose you are in Dr. Khan's position. Write a program to find all possible permutations of a given three-letter string S .

Note: You must use iteration to accomplish this.

Constraints

- $\text{len}(S) = 3$ (length of string is 3)
- All three letters of the string are unique.

Interaction

The input comprises a single 3-letter string S .

The output must be a single line containing all permutations of the three letters separated by a space.

Sample

Input	Output
"ABC"	ABC ACB BAC BCA CAB CBA
"XYA"	XYA XAY YXA YAX AXY AYX

In the first case, $S = \text{"ABC"}$. The 6 possible permutations are ABC, ACB, BAC, BCA, CAB, and CBA.

In the second case, $S = \text{"XYA"}$. The 6 possible permutations are XYA, XAY, YXA, YAX, AXY, and AYX.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
"NOM"	NOM NMO ONM OMN MNO MON
"APS"	APS ASP PAS PSA SAP SPA
"HUM"	HUM HMU UHM UMH MHU MUH

Pseudocode

```
S="ABC"
for i in S:
  for j in S:
    if j!=i:
      for k in S:
        if k!=j and k!=i:
          print(f"{i}{j}{k}",end=" ")
```

Dry Run

Below, dry run your pseudocode on any **one** of the inputs provided in the Exercise section.

Input: NOM

<pre> for i in S: (i=N) for j in S: (j=N) if j!=i: (False) for j in S: (j=O) if j!=i: (True) for k in S: (k=N) if k!=j and k!=i:(False) print("NOM") for i in S: (i=O) for j in S: (j=N) if j!=i: (True) for k in S: (k=N) if k!=j and k!=i:(False) for k in S: (k=O) if k!=j and k!=i:(False) for k in S: (k=M) if k!=j and k!=i:(True) print("ONM") </pre>	<pre> for i in S: (i=M) for j in S: (j=N) if j!=i: (true) for k in S: (k=N) if k!=i and k!=j: (false) for k in S: (k=O) if k!=j and k!=i:(true) print("MNO") for i in S: (i=O) for j in S: (j=M) if j!=i: (true) for k in S: (k=N) if k!=i and k!=j: (true) print("OMN") </pre>	<pre> for i in S: (i=N) for j in S: (j=M) if j!=i: (true) for k in S: (k=O) if k!=i and k!=j: (true) print("NMO") for i in S: (i=M) for j in S: (j=O) if j!=i: (true) for k in S: (k=N) if k!=i and k!=j: (true) print("MON") </pre>
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4. Introduction to Hackerrank [6.25 Points]

The guide to login/sign-up on Hackerrank tests is available [here](#).

Problems are available on this link as a short introduction to programming:

<https://hr.gs/cs101-lw5-th-fall24>